

Final Report :
Simulating 3D Effects of a Diamond-Shaped Airfoil

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1 Introduction

2 Theoretical Background: Oblique Shock Theory

3 Mesh Setup

The mesh of the diamond-shaped wing is generated from *OpenFOAM's* `blockMeshDict` which is set up as follows:

3.1 2D

In $\mathbb{R}^2$:

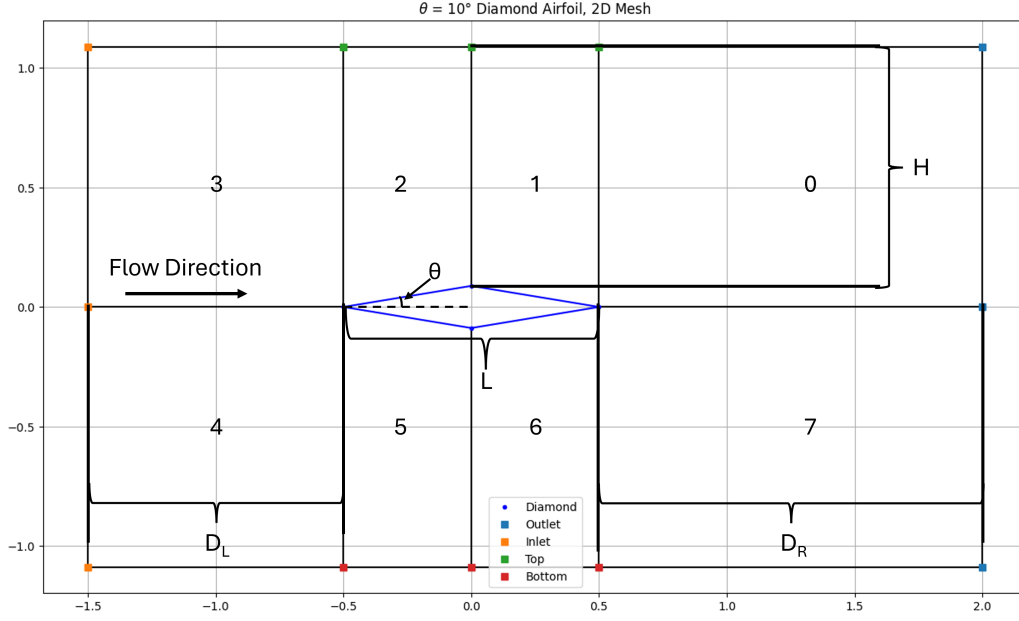


Figure 1: Vertices, Blocks, and Faces of the Diamond Wing Mesh

Where for our case:

- $L = 1.0$ (Chord)
- $\theta = 10^\circ$ (Deflection Angle)
- $D_L = 1.0$ (Distance Left of the Diamond - Fore)
- $D_R = 1.5$ (Distance Right of the Diamond - Aft)
- $H = 1.0$ (Distance Above/Below the top/bottom-most point of the Diamond)

The size of the computational domain is then $(D_L + L + D_R) \times (2H + L \tan(\theta)) \times 0.1 = 3.5 \times 2.176 \times 0.1$. The `blockMeshDict` for the $\mathbb{R}^2$ case can be found in the **Appendix**.

4 Mesh Refinement and Comparison to OST

5 Conclusion

Appendix

A Code

PDF of code starts on next page.

Additionally, team repository: <https://github.com/andressuniaga/CFD-SP25-11>

Order:

- `diamondmesh.py`
- `writeblockmeshDict2D.py`
- `writeblockmeshDict3D.py`
- $\mathbb{R}^2$ `blockMeshDict`
- $\mathbb{R}^3$ `blockMeshDict`